Fuel Type Analysis & US Market Entry Strategy

📌 1) Business Problem

Geely Auto plans to enter the highly competitive US automobile market by establishing a local manufacturing presence. To compete effectively with established US and European manufacturers, the company must understand customer preferences related to fuel type selection (Gas vs. Diesel).

Fuel type significantly influences vehicle design, engine configuration, pricing strategy, and market positioning. Without a clear understanding of the factors driving fuel choice, product development decisions may not align with US consumer demand.

Therefore, the key business problem is:

What vehicle characteristics influence customers' preference for gas or diesel cars in the US market?

🎯 2) Business Objective

The primary business objective is to:

- Identify the key technical and economic factors influencing fuel type selection.
- Understand how vehicle specifications impact customer fuel preference.
- Provide strategic recommendations for Geely Auto's product portfolio and market positioning.

🧠 3) Project Objective (Technical Objective)

The objective of this project is to build a **Logistic Regression model** that predicts whether a vehicle uses:

- Gas (0)
- Diesel (1)

based on independent variables such as engine size, compression ratio, horsepower, fuel efficiency, and price.

The model will help quantify the relationship between vehicle attributes and fuel type selection.

🛠 4) Analytical Approach

The project was executed using a structured data science workflow:

Step 1: Data Understanding & Cleaning

- Loaded dataset
- Checked for missing values and duplicates
- Dropped irrelevant columns (car_ID, CarName)

Step 2: Exploratory Data Analysis (EDA)

- Univariate analysis of numerical and categorical variables
- Bivariate analysis between fuel type and key features

- Identified class imbalance in target variable

Step 3: Data Preprocessing

- Encoded categorical variables using one-hot encoding
- Performed train-test split
- Applied SMOTE to handle class imbalance
- Scaled numerical variables using StandardScaler

Step 4: Model Building

- Trained Logistic Regression model on balanced dataset

Step 5: Model Evaluation

- Accuracy Score
- Confusion Matrix
- Classification Report
- ROC Curve and AUC Score

📊 5) Evaluation Metrics Used

To assess model performance, the following metrics were used:

🔹 **Accuracy**

Measures overall correctness of predictions.

🔹 **Precision**

Measures how many predicted diesel cars were actually diesel.

🔹 **Recall**

Measures how well the model identifies actual diesel vehicles.

🔹 **F1-Score**

Balances precision and recall.

🔹 **ROC-AUC**

Measures overall classification performance independent of threshold.

Since the dataset was imbalanced, special focus was placed on Recall and ROC-AUC rather than only accuracy.

🔍 6) Key Assumptions

- Vehicle specifications strongly influence fuel type selection.
- Historical vehicle data reflects general US market behavior.
- SMOTE-generated samples represent realistic minority class patterns.

💼 7) Business Impact

The model provides measurable insights into:

- Engine configurations associated with diesel vehicles
- Performance attributes influencing premium segments
- Fuel efficiency preferences among US customers
- Price positioning strategies

This enables Geely Auto to:

- Design vehicles aligned with US demand
- Optimize product portfolio mix
- Reduce market entry risk
- Improve competitive positioning

🚀 8) Strategic Recommendation Summary

Geely Auto should adopt a dual strategy:

1. Prioritize gas vehicles for mass-market penetration.
2. Introduce selective diesel variants in mid-to-premium SUV segments.
3. Emphasize performance, durability, and fuel efficiency in marketing communication.
4. Align pricing with customer value perception.

🏁 9) Final Conclusion

The analysis successfully identified significant predictors of fuel type selection in the US market. Logistic Regression modeling combined with SMOTE balancing provided reliable classification performance and interpretable insights.

The results support data-driven decision-making for Geely Auto's US market entry strategy.